

This password follows every rule.

Password123!

- ✓ Uppercase
- ✓ Lowercase
- ✓ Number
- ✓ Symbol
- ✓ 12 characters

38 WEAK
out of 100 in my analyzer

Here's why → swipe

WHAT I BUILT

An analyzer that thinks like an **attacker**, not a checklist.

Each character becomes a tile, coloured by the pattern it belongs to.

ANALYZER

Password

Password123!

 Hide password

Clear

Results update as you type. Hidden by default so people nearby can't read it.

► Optional: check against personal details (demo values only)

WEAK

38/100

VERY WEAK

WEAK

MODERATE

STRONG

VERY STRONG

PATTERN MAP · WHERE THE PREDICTABILITY IS

P

a

s

s

w

o

r

d

1

2

3

!

Sequence

Dictionary word

✓ Good length (12+ characters)

✓ Not a common password

✓ No keyboard patterns

⚠ Word + number structure

✓ Character variety (3+ types)

⚠ Predictable sequence

✓ No repeated patterns

✓ No years or dates

PASSWORD STRENGTH ANALYZER

02 / 08

They don't guess randomly. They try **patterns** first.

Common passwords

123456

Keyboard walks

qwerty · asdf

Sequences

1234 · abcd

Repetition

aaaa · abcabc

Word + number / year

Welcome2026!

Personal details

Name@1999

My tool detects all six (plus leetspeak like p@ssw0rd) and explains each one.

The textbook formula says it's strong. **It isn't.**

78.8

bits: $L \times \log_2(N)$

Assumes every character was picked at random.

23.0

bits: pattern-adjusted

Re-prices "Password", "123" and "!" at what they cost an attacker.

Predictable structure HIGH

A common word followed by digits - a pattern guessing tool applies to every word.

Predictable sequence MEDIUM

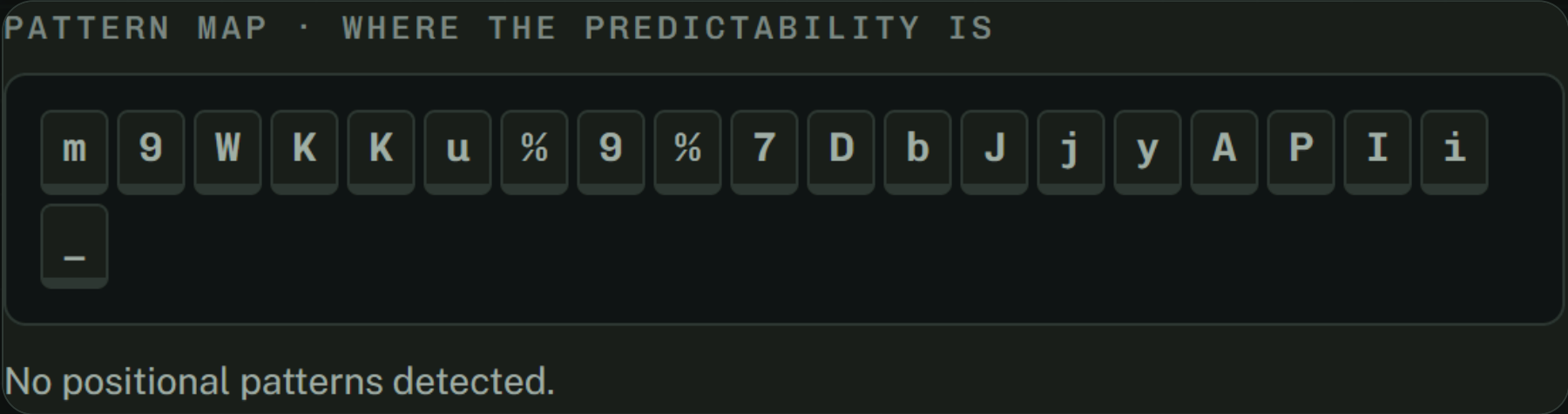
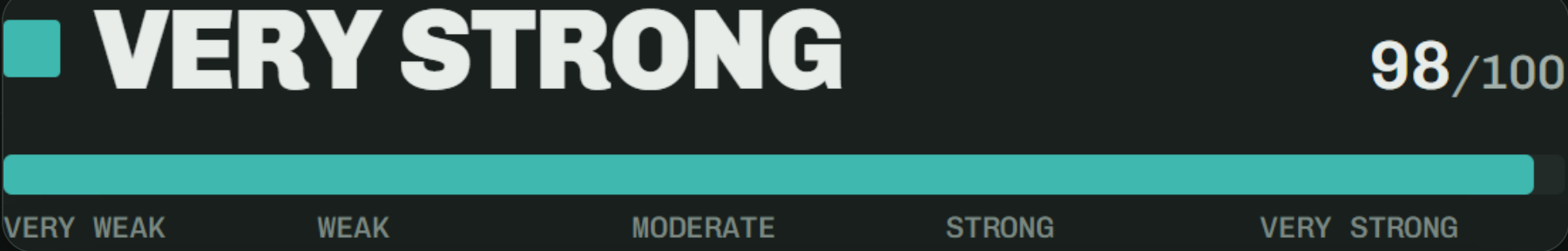
Found 1 sequential run of characters (longest: 3 characters), like 1234 or dcba.

Dictionary word MEDIUM

Contains 1 common word.

Length + real randomness wins.

A 20-character password from the built-in generator (Python secrets, a CSPRNG) scores 98/100 with zero patterns.



A password tool must never become a password **leak**.

- ✓ Analyzed in memory: never stored, logged or sent to third parties
- ✓ No password column in the database, not even a hash
- ✓ Never in URLs, browser storage, API responses or error messages
- ✓ 73 automated tests, including ones that scan the raw database file and logs

73/73

automated tests passing

Including 12 dedicated privacy & security tests.

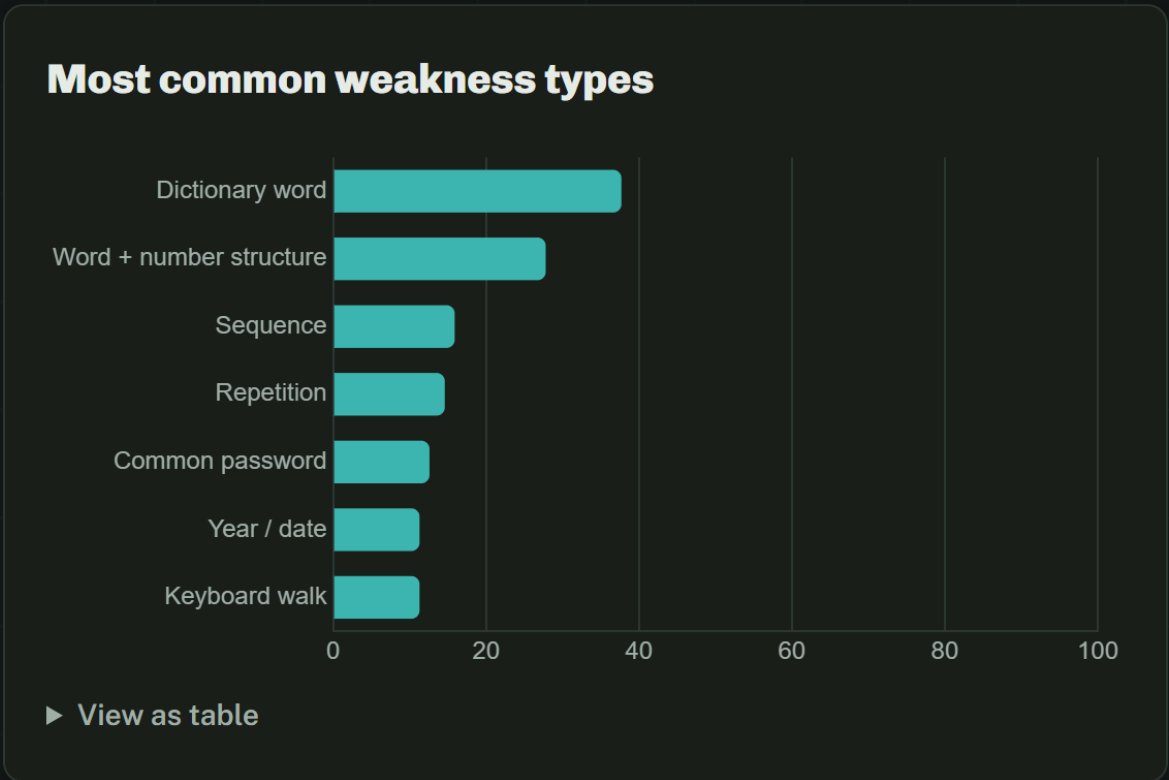
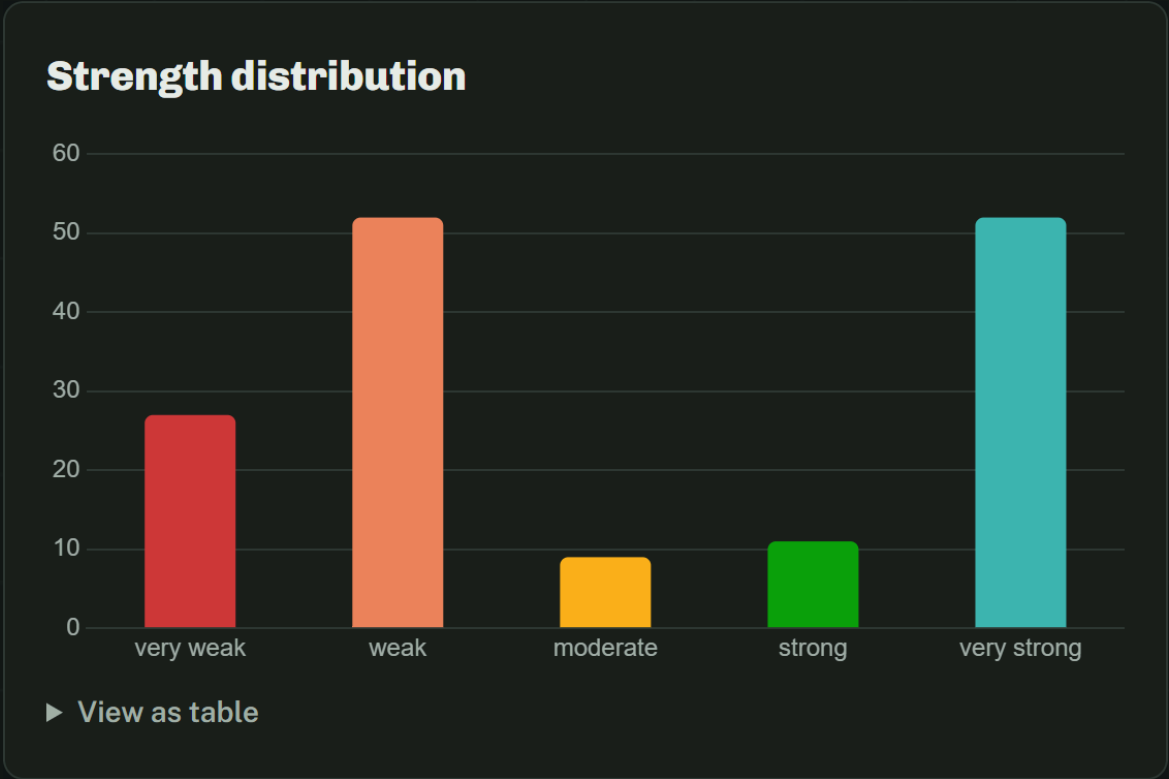
0

password columns

Only score, class, length and weakness types, and only on opt-in.

A dashboard that only sees metadata.

Score, class, length and weakness types, recorded only when a user opts in. (Synthetic demo data shown.)



Three habits beat any clever password trick.

1

Unique password per account

Reuse is how credential stuffing works.

2

Use a password manager

Let it generate long, random passwords for you.

3

Turn on MFA

A strong password doesn't stop phishing. A second factor does.

Built with Python · Flask · JavaScript · SQLite

Full source code on GitHub. Link in bio / comments.